LIVEWIRE™
FOR LIVE CAREERS

Machine Learning using python- Day wise Syllabus

Day	Topics 4hours/day
1	Introduction to Machine Learning, History, Applications, Introductory concept of python, Data structures ,Control flow, Functions
2	Modules and packages, File input and output, Exception handling, Standard libraries, Preliminaries of Statistics
3	Statistic Data types, Measure of Central Tendency & Dispersion, Probability Theory, Normal Distribution, case study, Scatter Diagram
4	Correlation Analysis case study, Confidence Interval, T-distribution, Case Study, Confusion matrix, Precision, Recall, F-score
5	Pre Processing, Classification algorithm: Decision Tree, GINI, case study
6	Entropy, KNN, SVM, Linear classifier: Naïve Bayes Classifier, Case study
7	Logistic regression, Regression Analysis, Linear Regression, Multiple Linear Regression, Case study
8	Random Forest, Poisson Regression, Lasso & Ridge Regression, Zero-Inflated Regression, Negative binomial regression
9	Multinomial Regression, Survival Analysis, Web Extraction, Case study, Clustering: Euclidean distance, Manhattan Distance
10	Hierarchical clustering, k-means clustering, k-Medoids, DBSCAN, PAM, CLARA, Case Study
11	Dimension Reduction, PCA, Word cloud, , Introduction to Text Mining, case study
12	Data Extraction and NLP, Introduction to Deep learning, Introduction to neural network, ANN for classification, case study
13	ANN for regression, Case study, Introduction to RNN, Difference between ANN and RNN, Methods and functions of RNN
14	RNN algorithm writing, RNN Case study, Projects